Potential of the Amazonian liana **(Cissus verticillata)** for a **sustainable food industry.**

Author: *Huaman Ñahui, Richard Ivan, rhuamann05@unia.edu.pe.*
Advisor: *Iván Ochoa-Moreno. University of York, York, United Kingdom.*
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Background

Plant-based hydrocolloids are essential in the food industry to enhance the viscosity and stability of products such as jams and dressings while also providing soluble fiber. However, the local market relies on imported additives like carboxymethyl cellulose (CMC), which is costly to produce and requires petroleum-derived chemical reagents. This dependency drives up production costs, creates risks of supply shortages, and increases environmental impact. The Amazonian climbing plant known as "bejuco ubí" (*Cissus verticillata*) offers a sustainable and significantly more affordable alternative. A lack of systematic research into its mucilage limits its potential for technological application as a cost-effective additive.

Research objectives

General Objective

To determine the potential of *Cissus verticillata* stems as a source of hydrocolloids using ultrasound-assisted acid extraction.

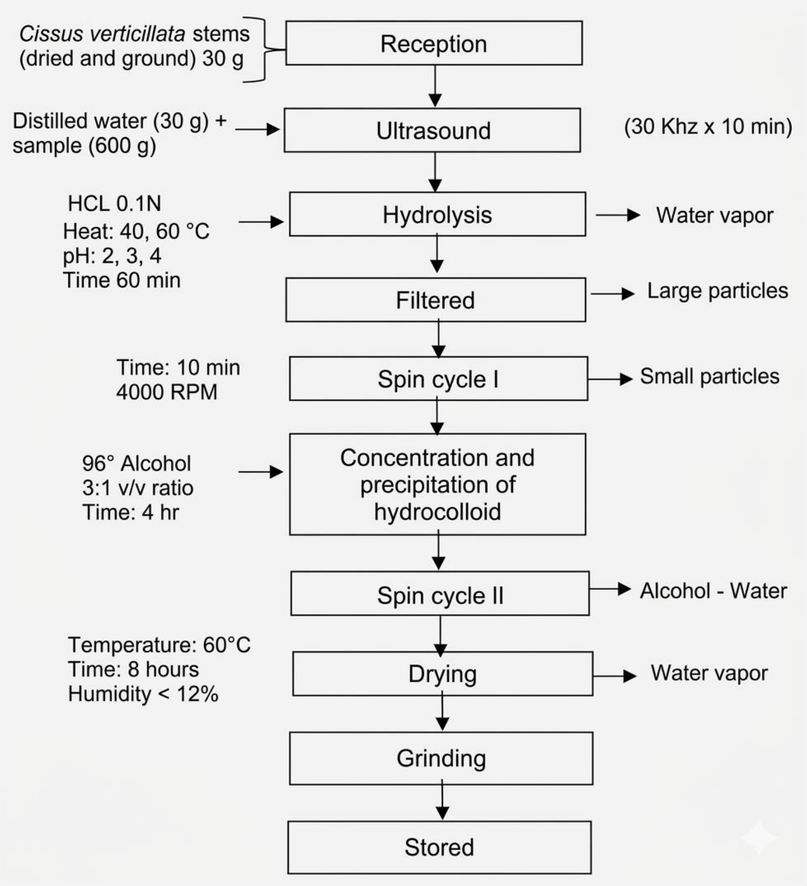
Specific Objectives

- Evaluate the extraction yield of mucilage from *Cissus verticillata* stems by analyzing the combined effect of pH and temperature using an ultrasound-assisted method.
- Identify the optimal treatment to maximize the recovery of this polymeric matrix.

Methodology

Cissus verticillata stems were collected as raw material from the Lukas farm, located in Pucallpa, Peru. A 2 × 3 factorial design was applied, evaluating two temperatures (40 and 60 °C) and three pH levels (2, 3, and 4), resulting in a total of six treatments. Extraction was performed using an ultrasound-assisted acid treatment. The recovered fraction was evaluated based on extraction yield and its characteristics to determine its potential as a hydrocolloid. The results were subjected to statistical analysis to compare the treatments.

Figure 1. Flowchart for the extraction of hydrocolloid from the stems of bejuco ubí (*Cissus verticillata*).



The extraction yield of the hydrocolloid was calculated using the following equation:

$$performance\ (\%) = \left(\frac{Mass\ of\ the\ dry\ hydrocolloid\ (g)}{Mass\ of\ the\ dry\ raw\ material\ (g)}\right) x\ 100$$

Results

Extraction Performance Analysis

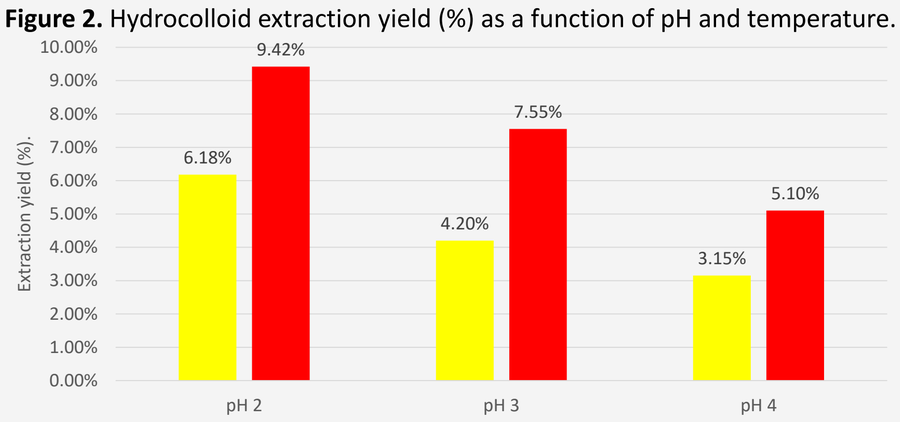
Analysis of variance (ANOVA) demonstrated that pH and temperature, as well as their interaction, have a statistically significant effect ($p < 0.05$) on yield.

The specific quantitative yields and macroscopic characteristics observed for each of the six experimental treatments are presented in detail in **Table 1** and **Figure 2.**

Table 1. Hydrocolloid extraction yield (%) as a function of pH and temperature.

Treatment	pH	Temperature (°C)	Performance (%)	Appearance and Visual Characteristics
T1	2	40	6.18%	Lumpy powder, light brown color, moderate solubility
T2	3	40	4.20%	Fine powder, uniform cream color, low viscosity
T3	4	40	3.15%	Fine powder, off-white color, low gelling capacity
T4	2	60	9.42%	Light brown semi-compact powder, high apparent viscosity
T5	3	60	7.55%	Hygroscopic, cream-colored powder with good swelling capacity.
T6	4	60	5.10%	Fine powder, opaque white color, high solubility

Figure 2. Hydrocolloid extraction yield (%) as a function of pH and temperature.



Treatment 4 (pH 2 and 60 °C) yielded the highest recovery, with 9.42% of the mass recovered. This demonstrates that more acidic conditions, combined with higher temperatures, favor the solubilization and depolymerization of the liana polysaccharides—processes that are optimized by ultrasonic waves. Conversely, Treatment 3 (pH 4 and 40 °C) showed the lowest yield, at only 3.15%, due to insufficient acid hydrolysis.

Conclusions

- Results confirmed that the material extracted from *Cissus verticillata* stems possesses the physical characteristics of a hydrocolloid, validating this species as a local, sustainable source capable of replacing imported additives in food technology.
- Mucilage extraction is highly sensitive to the experimental environment; ANOVA confirmed that pH, temperature, and their interaction exert a statistically significant effect ($p < 0.05$) on the physical yield.
- Treatment 4 (pH 2 and 60 °C) optimized the depolymerization and solubilization of the polysaccharides through the synergistic action of ultrasound, achieving a maximum yield of 9.42% recovered mass.
- Insufficient acidity and heat limit the process, with the lowest recovery observed in Treatment 3 (pH 4 and 40 °C)—at just 3.15%—due to poor hydrolysis.
- Future research should focus on the detailed rheological characterization and purification of the extract to determine its specific industrial applications.

Literature

Scan the QR code to access the complete list of bibliographic references (Vancouver style).